Baseball Salaries and Performance, 2000–2016

Exploratory Data Analysis | COP 3045 | Team 5

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1. Purpose and dataset

This report examines salary distributions, relationships between salary and batting statistics, differences between leagues and teams, and the highest recorded player salaries.

The cleaned dataset contains 18,649 rows, 37 columns, and 3,359 unique players across 2000–2016. All analyses use data/clean_data.csv.

2. Data preparation

The cleaning script removes exact duplicate rows and converts numerical fields to numbers. Invalid numerical entries become missing values rather than being replaced with averages. The graphical analysis also trims whitespace from categorical values.

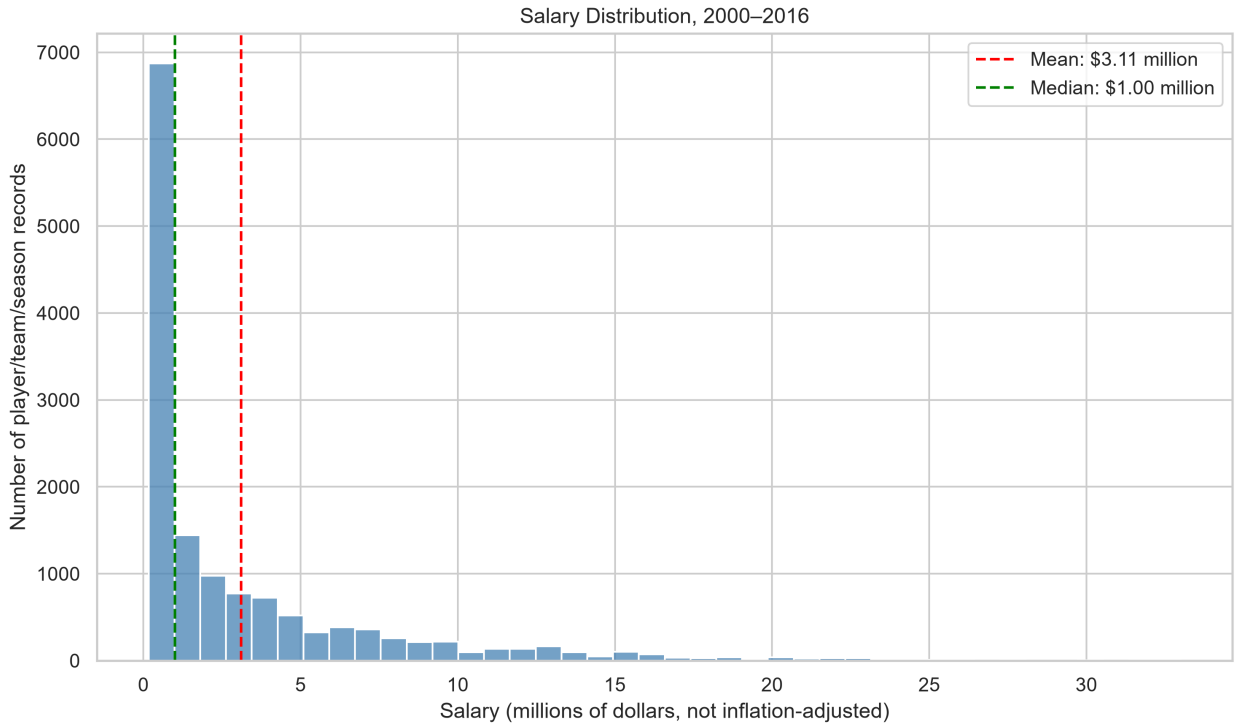
Because salaries repeat across position rows, salary analyses use one record per player, team, and season, giving 14,165 salary records. Batting analyses remove repeated copies of batting stints and sum separate stints within each player/team/season before joining salary.

3. Missing data

salary: 0 (0.00%)
AB: 517 (2.77%)
POS: 579 (3.10%)
PB: 17,491 (93.79%)
WP: 18,649 (100.00%)
ZR: 18,649 (100.00%)

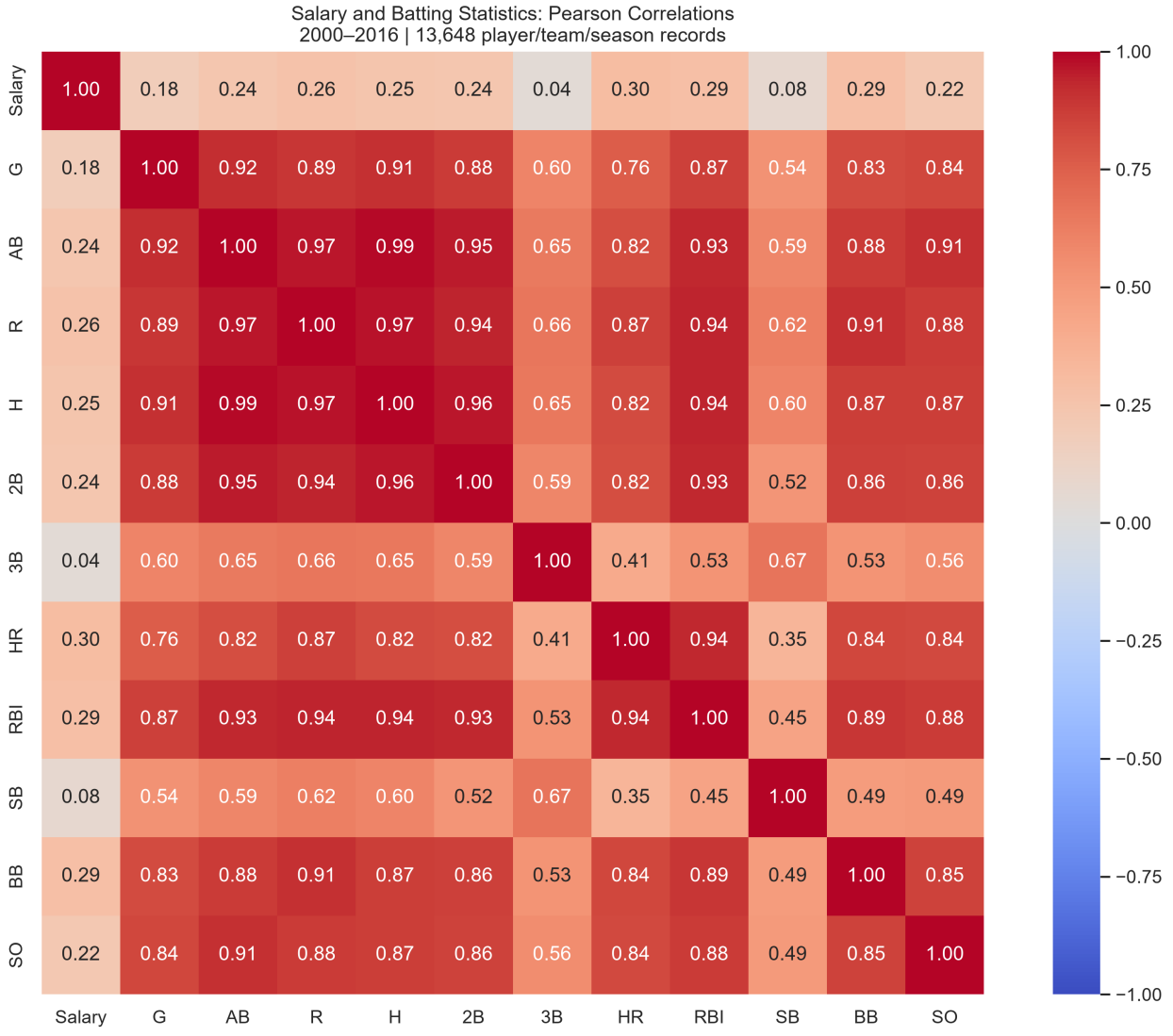
The correlation analysis uses 13,648 complete player/team/season records with the selected batting variables. Missingness varies substantially by column; retaining missing values does not guarantee unbiased results.

Figure 1. Salary distribution



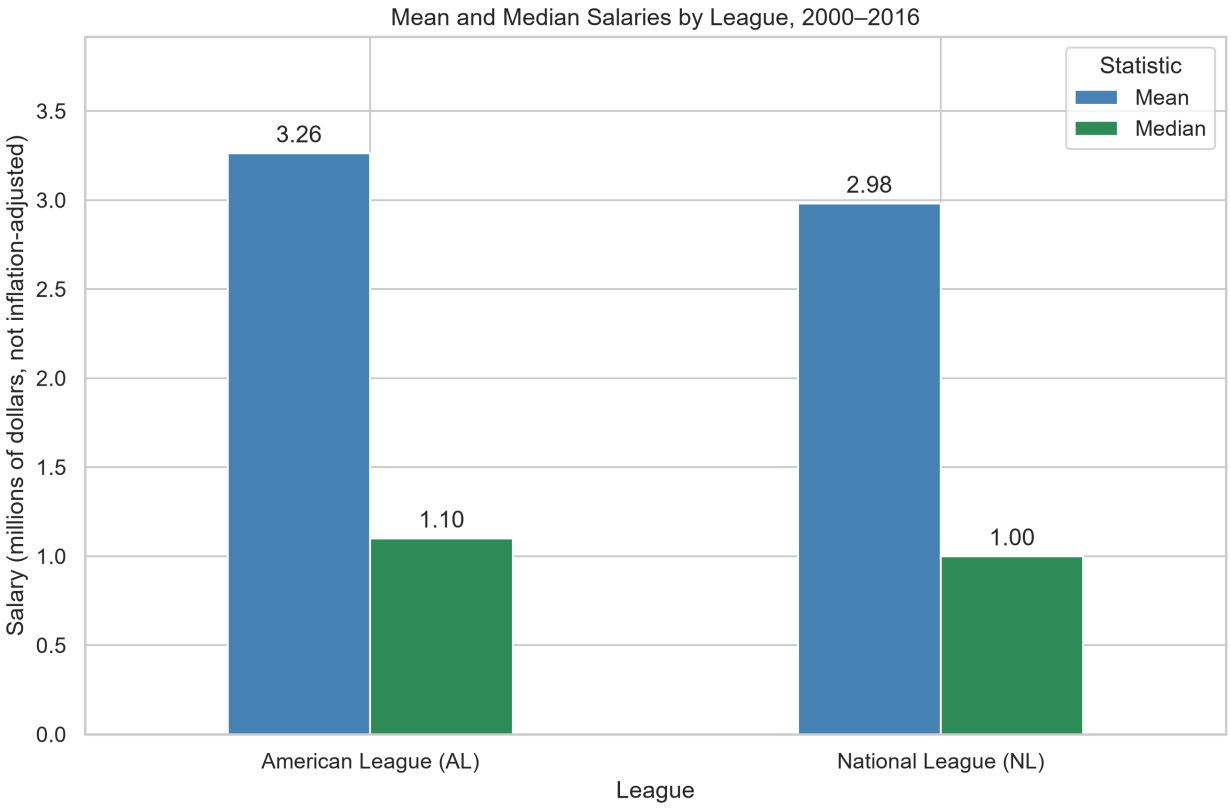
The mean salary is $3,114,436.59, compared with a median of $1,000,000.00. Salary skewness is 2.30. The positive skew and higher mean indicate that a relatively small number of high salaries pull the average upward. The median is less affected by these extreme values. Salaries are pooled across seasons and are not inflation-adjusted.

Figure 2. Salary and batting correlations



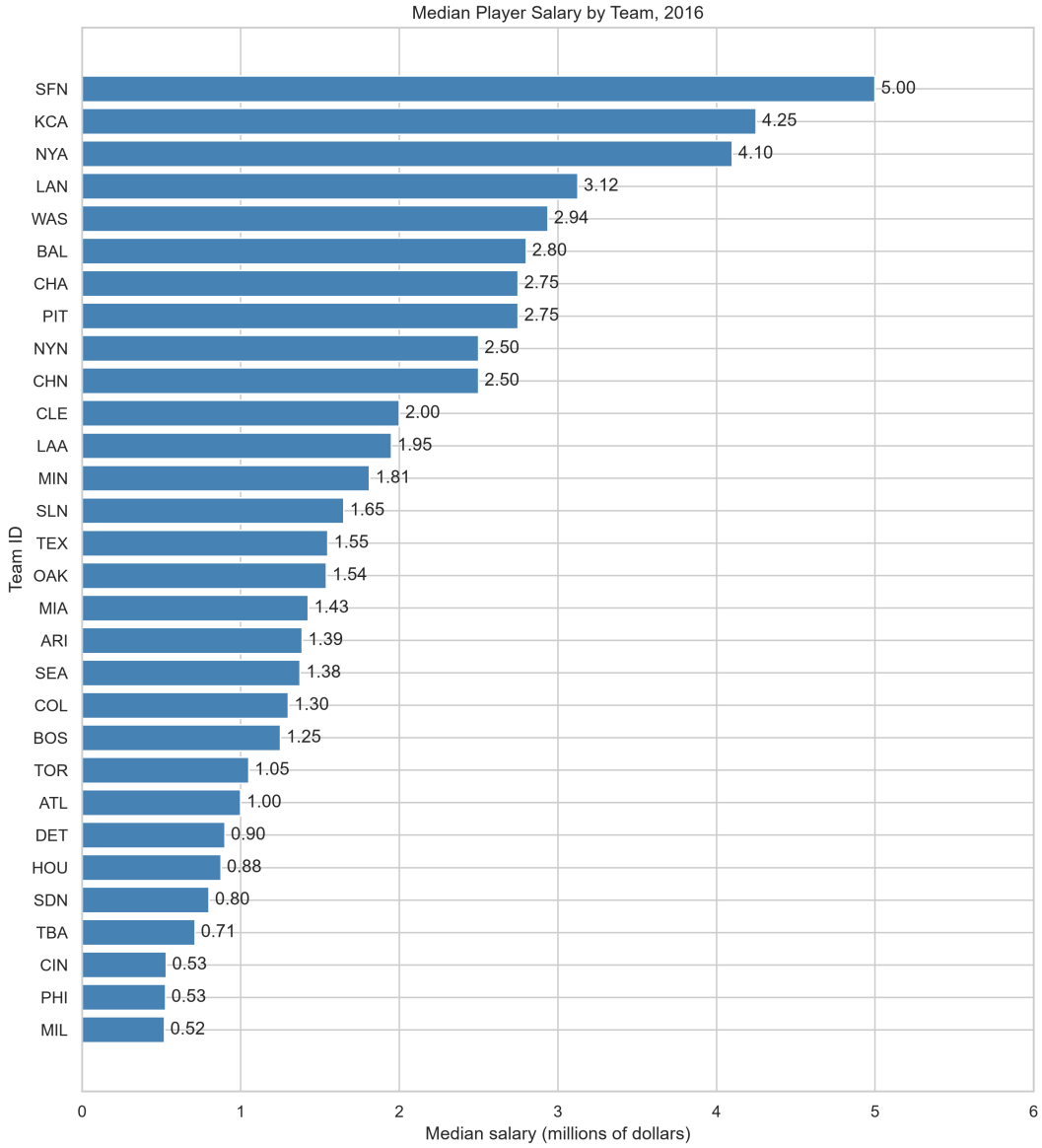
Among the selected batting statistics, HR has the largest absolute Pearson correlation with salary (r = 0.301). These correlations describe linear associations, not causation. The analysis pools seasons and includes pitchers with batting records, without adjusting for position, playing time, or season. G = games; AB = at-bats; R = runs; H = hits; 2B = doubles; 3B = triples; HR = home runs; RBI = runs batted in; SB = stolen bases; BB = walks; SO = strikeouts.

Figure 3. Salary comparison by league



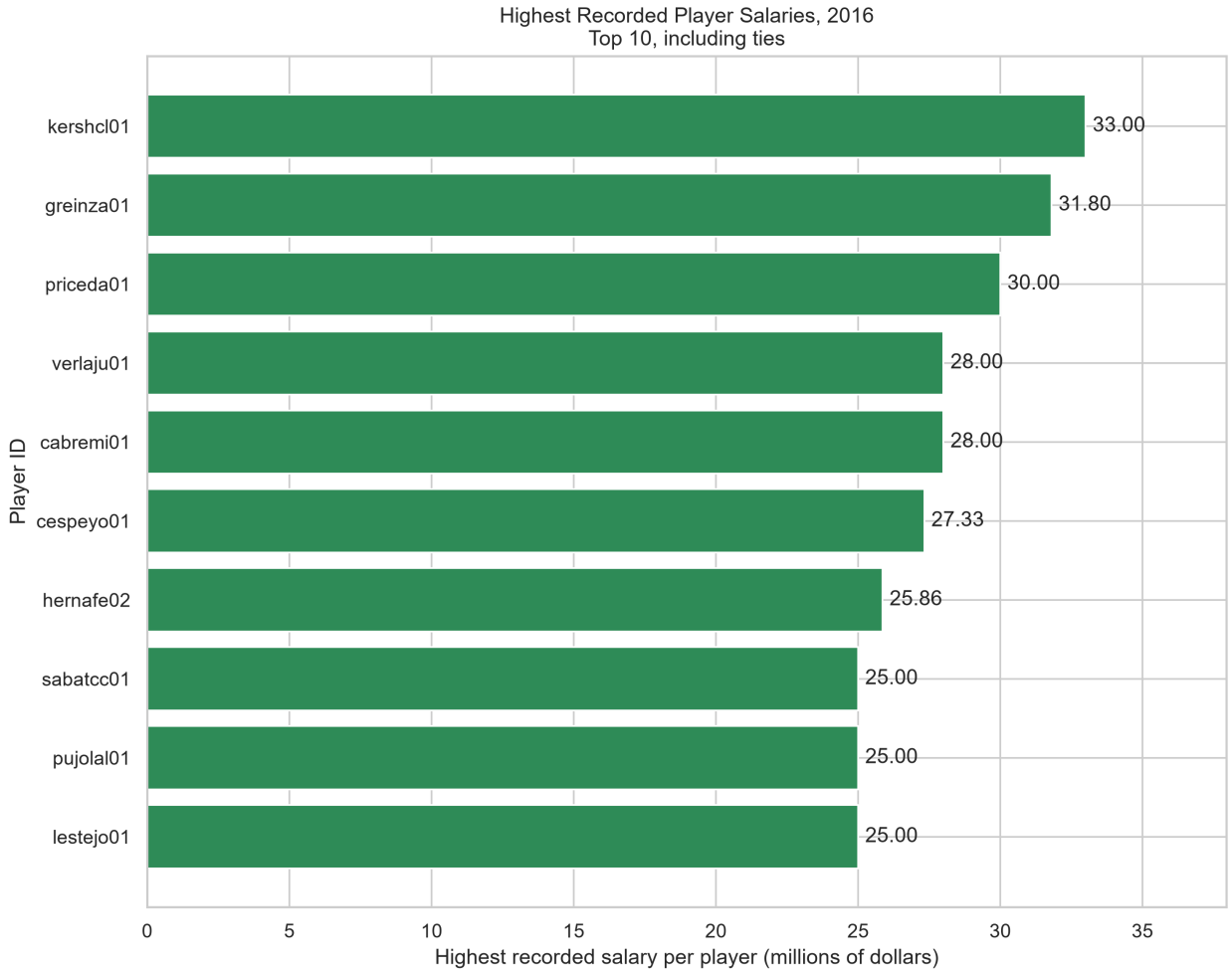
AL: mean $3,263,345.59; median $1,100,000.00; 6,708 player/team/season records. NL: mean $2,980,484.38; median $1,000,000.00; 7,457 player/team/season records. The median describes typical salaries, while the mean is more affected by high earners. This pooled comparison does not adjust for season or player composition.

Figure 4. Median salary by team, 2016



SFN has the highest median in this dataset at $5,000,000.00, while MIL has the lowest at $518,300.00. The comparison uses a single season and measures typical player salary, not total team payroll. Team labels are dataset identifiers.

Figure 5. Highest recorded player salaries, 2016



Player kershcl01 has the highest recorded salary in the ranking at $33,000,000.00. The chart includes ties at the tenth-highest salary. For players with records for multiple teams, the highest recorded salary is used rather than adding potentially overlapping amounts. These values are not career earnings. Labels are player IDs.

4. Limitations

Results describe the supplied dataset and may not represent every player or every component of compensation. Missing performance records may affect the correlation sample. Position-specific statistics should not be treated as universally applicable.

Repeated observations of the same player across seasons remain in the pooled analyses. Salaries are not adjusted for inflation. The analysis does not control for experience, contract terms, position, or other potential influences on pay. No causal or statistical-significance claims are made.

5. Conclusion

The salary distribution has a mean of $3,114,437 and a median of $1,000,000, illustrating the influence of high earners on the average. League and team comparisons describe differences in salary levels. The batting correlations provide an initial view of performance associations, but should be interpreted alongside the study's limitations.

Reproducibility

Run 04-EDA/generate_report.py after data/clean_data.csv has been created. The report script runs graphical_eda.py, regenerates all five figures, calculates the report values, and writes this PDF without manual image insertion.